CERTIFICATES FOR PURE CUBIC THUE EQUATIONS: MERLIN–ARTHUR MEMBERSHIP AND THE COMPACT VANISHING PROBLEM

THE DIOPHANTINE-FABLE EXPEDITION

Abstract. For the pure cubic Thue equation $x^3 - d\,y^3 = m$, the companion paper gave a deterministic $2^{O(s)}$ decision algorithm. Here we ask what *certifies* solvability. The certificate collapses to a single object: a Thiel-format compact representation of the solution element $\beta = x - y\alpha$ itself. Integrality and the norm equation $N(\beta) = m$ verify deterministically in polynomial time — the latter by multiplicativity of norms on power products, the mechanism behind Lagarias's Pell certificates and Ge's equality tests. Membership in **MA** follows unconditionally. Membership in **NP** is equivalent, modulo this architecture, to one newly isolated problem — the **Compact Vanishing Problem** (CVP): decide deterministically whether a fixed coordinate functional vanishes at a compactly represented algebraic integer. We classify CVP into the sum-of-square-roots family, prove that ideal-theoretic and radical-rigidity methods cannot express it, and explain by a vacuity remark why Lagarias's degree-2 certificates never met it.

1. Introduction

1.1. **From decision to certification.** The companion paper [companion] proved that pure cubic Thue equations $x^3 - d\,y^3 = m$ ($d \geq 2$ cube-free, not a cube; $m \neq 0$; input size $s = O(\log d + \log |m|)$ in binary) are decided, with full solution lists, by a deterministic unconditional algorithm in time $2^{O(s)}$ — a first worst-case bound of this shape for a Thue stratum of Smale's fifth problem [Sma98]. Its Remark 7.5 named the three obstructions to *polynomial-time* decision: the unit (the principal ideal problem, classically believed hard, polynomial only quantumly [Hal07]); the zero test (certified zero-testing against $2^{O(s)}$-digit height bounds); and the window (quasi-linear in $R = 2^{\Theta(s)}$ ([companion, Lemma C], after Sha [Sha19]) unless the strong window conjecture is proven).

The natural target below polynomial time is therefore **NP membership**: poly-size certificates, deterministic poly-time verification. This paper is that

Date: Sequel draft of 2026-07-22, assembled from the certificate notes; a companion to the exponential-time paper [companion], whose notation it reuses. Every honesty flag of the notes is carried forward: sketches are labeled, quoted-standard facts are flagged, the status ledger is reproduced in Section 7, and no claim below exceeds what the underlying notes established.

Draft — authorship placeholder.

pivot, and its outcome is asymmetric: two thirds of the verifier's work is classical and easy, and the entire remaining difficulty isolates into one new problem.

1.2. **One rung down: Lagarias's Pell certificates.** Lagarias [Lag79] proved that solvability of binary quadratic Diophantine equations over $\mathbb{Z}$ is in NP, although the fundamental solutions have exponentially many digits: the certificate is a power-product representation, and both verifiable conditions — integrality and the norm equation — are *multiplicative*, hence compatible with power products. (Negative Pell decision is even in NP $\cap$ coNP by the same technology, with no classical polynomial algorithm known.) This paper asks what happens to that magic one rung up, at degree 3.

1.3. **The two main results, and the crux they isolate.** The certificate architecture (Section 2) simplifies an outside reviewer's proposed three-part scheme (unit + reduced representative + orbit index): all three collapse into the single object β, the solution element itself in compact representation. Verification splits into three steps — integrality, norm, Thue shape — and:

- **Integrality and norm verify deterministically in polynomial time** (Section 2), by the format's denominator bookkeeping and by norm multiplicativity on power products, the latter a special case of Ge's deterministic multiplicative-relation testing [Ge93].
- **Proposition 3.1 (MA membership, unconditional).** Pure cubic Thue solvability is in MA, with one-sided error: the remaining condition — the vanishing of one coordinate — is testable modulo random primes, with the prime range calibrated against an explicit height bound extracted from the compact data.
- **The crux (Section 4).** The single non-derandomized step is named: **CVP, the Compact Vanishing Problem** — decide deterministically whether a fixed $\mathbb{Q}$-linear coordinate functional vanishes at a compactly represented algebraic integer. Cubic Thue $\in$ NP **iff-modulo** CVP (Corollary 4.1, with the honest gloss given there); conditionally, under derandomization hypotheses implying MA $=$ NP, membership in NP holds today.
- **The vacuity remark (Remark 4.5).** At degree 2 the module $\mathbb{Z} + \mathbb{Z}\sqrt{d}$ is *all* of $\mathbb{Z}[\sqrt{d}]$: the shape condition is vacuous and CVP never arises. The coordinate condition is genuinely new at degree 3 — CVP, not the unit computation, is the honest frontier of the NP question.

Section 5 explains why CVP resists the classical toolbox: four attack routes — Thiel/Ge equality testing, Blömer's radical sums, ideal-theoretic reformulations, the Baker route — are each shown inapplicable, for one unified reason.

1.4. **Organization.** Section 2 fixes the certificate and verifier; Section 3 proves MA membership; Section 4 states CVP, its three faces, and the vacuity remark; Section 5 is the impossibility ledger; Sections 6–7 collect positive fragments, open problems, and the status ledger.

2. The certificate and the verifier

2.1. **Notation (from the companion).** $d = ab^2 \geq 2$ cube-free, not a cube, a, b coprime squarefree; $\alpha = \sqrt[3]{d}$; $K = \mathbb{Q}(\alpha)$, complex cubic, one real embedding σ_1 and a conjugate pair σ_2, σ_3; unit group $\langle -1, \varepsilon \rangle$ of rank one, regulator $R = \log \sigma_1(\varepsilon) = 2^{O(s)}$ [companion, § 2.3]. For $g \in K$ write $g = c_0(g) + c_1(g)\alpha + c_2(g)\alpha^2$; the index $[O_K : \mathbb{Z}[\alpha]]$ divides $3b$, so $3b\, c_i(g) \in \mathbb{Z}$ for $g \in O_K$ [companion, § 2.2]. A solution (x, y) of $x^3 - dy^3 = m$ is the element $\beta = x - y\alpha \in \mathbb{Z}[\alpha]$ with $N(\beta) = m$ and **Thue shape** $z(\beta) := c_2(\beta) = 0$. We write $M := \mathbb{Z} + \mathbb{Z}\alpha$ for the rank-2 module in which solutions live.

2.2. **Compact representations.** A **compact representation** (Thiel format) of $\beta \in O_K$ is a power product

$$\beta \;=\; \beta_0 \cdot \prod_{i=1}^{T} \beta_i^{2^i},$$

with $T = O(\log h(\beta))$ factors, each $\beta_i \in K$ given by poly(s) bits of coordinate and denominator data. The existence and size bounds are Thiel's, **unconditional and in arbitrary degree** [Thi95, Def. 6.3.1, Cor. 6.3.9–6.3.10]: every unit and every generator of a principal ideal admits a compact representation with $T \leq \log(\log n + (n-1)\log H) + 2 = O(\log h(\beta))$ factors of poly$(\log|\Delta|)$ bits each. (Thiel's normal form uses descending exponents 2^{T-i} and rational factors α_i/d_i, $d_i \leq \sqrt{\Delta}$ — cosmetically, not mathematically, different from the form displayed above; the leading factor additionally carries $O(\log|N(\beta)|)$ bits, harmless here.) Efficient computation from power-product data and log-embedding approximations is Biasse–Fieker [BF14, Alg. 7, Prop. 5.1], unconditional given its input. The quadratic antecedent is [BTW95].

For our instances a solution element has log-height $O(R + \log|m|) = 2^{O(s)}$ (unit reduction and the two-sided window of [companion, Lemmas U and C]), so $T = O(s)$: the certificate has poly(s) size. Existence is *constructive*: the honest prover runs the companion's exponential algorithm once; the tracked-generator walk of [companion, Lemma A′], compacted into Thiel's normal form via [BF14, Alg. 7], emits exactly this format. (Schoof's Arakelov machinery [Sch08] supplies the reduced-divisor arithmetic used elsewhere but contains no compact-representation theorem — a citation-scope point verified against the source.)

2.3. **The certificate is the solution element itself.**

Certificate (YES instances). A Thiel-format compact representation of $\beta = x - y\alpha$.

No ε, no reduced representative, no orbit index k, no window: if β is a solution, its compact representation alone witnesses it. In particular the strong window conjecture of [companion, Remark 7.4] is *not needed* for YES-certificates at all; it re-enters only for the prover-efficiency narrative and on the coNP side (Section 6.2).

2.4. The verifier, step by step.

(1) **V1 — Integrality**: $\beta \in O_K$, and $\beta \in \mathbb{Z}[\alpha]$ after the $3b$-bookkeeping. Deterministic polynomial time: the format carries denominator data checked by exact ideal arithmetic on poly-size objects; and since coordinate denominators in O_K divide $3b$, membership in $\mathbb{Z}[\alpha]$ is a *congruence* condition modulo $3b$, computable by repeated squaring in a poly(s)-size quotient ring. *(Mechanism sketch; the format-level details are Thiel's.)*

(2) **V2 — Norm**: $N(\beta) = m$. Deterministic polynomial time, because **norms are multiplicative on power products**:

$$N(\beta) \;=\; N(\beta_0) \cdot \prod_{i=1}^{T} N(\beta_i)^{2^i},$$

each $N(\beta_i)$ a small explicit rational, exponents in binary — exactly the magic that powered Lagarias's Pell certificates [Lag79]. Verifying the exponent identity without factoring is standard multiplicative technology: coprime-basis bookkeeping [vG25], or a special case of Ge's multiplicative-relation test [Ge93]; the sign is read off certified embedding data. *(Sketch; the point is that every ingredient is multiplicative.)*

(3) **V3 — Thue shape**: $z(\beta) = 0$. **This is the entire remaining difficulty.** It is testable *randomly* in polynomial time (Section 3); whether *deterministically* is precisely CVP (Section 4).

Completeness of the architecture is immediate: $\beta \in \mathbb{Z}[\alpha]$ with $N(\beta) = m$ and $c_2(\beta) = 0$ *is* a solution $(x, y) = (c_0(\beta), -c_1(\beta))$, and conversely.

3. Merlin–Arthur membership

Proposition 3.1 (MA membership, unconditional). *Pure cubic Thue solvability is in MA, with perfect completeness and one-sided, amplifiable soundness error.*

Proof (complete modulo the compact-representation format bounds quoted in Section 2.2*).* Merlin sends the compact representation of a claimed solution element β. Arthur runs V1 and V2 of Section 2.4 deterministically, then tests V3 modulo random primes.

The integer to test. Since $\alpha, \beta \in O_K$, the trace $W := \mathrm{Tr}_{K/\mathbb{Q}}(\beta\alpha)$ is a rational integer, and by the trace identity of Section 4.2,

$$z(\beta) = 0 \iff W = 0.$$

Height bound from the compact data. Embeddings are multiplicative on the format: for each j,

$$\log|\sigma_j(\beta)| \;=\; \log|\sigma_j(\beta_0)| + \sum_{i=1}^{T} 2^i \log|\sigma_j(\beta_i)|.$$

Each factor carries $\mathrm{poly}(s)$ bits, so $\log|\sigma_j(\beta_i)| \le P(s)$ for an explicit polynomial P, and $T = O(s)$ gives $\log_2 \mathrm{house}(\beta) \le P(s)\, 2^{T+1} \le 2^{c_3 s}$ for an explicit c_3. Hence $|W| \le 3\,\mathrm{house}(\beta)\,\mathrm{house}(\alpha)$, so $\log_2|W| \le 2^{c_3 s} + O(s) \le 2^{c_4 s}$; if $W \ne 0$, its number of distinct prime divisors obeys $\omega(W) \le \log_2|W| \le 2^{c_4 s}$.

The prime range for a $2/3$ threshold. Let E be the set of primes dividing $3d$ or any denominator in the format; $\#E \le \mathrm{poly}(s)$. Arthur draws p uniformly from the primes in $[17, 2^t]$: by Chebyshev-type bounds ($\pi(x) > x/\ln x$ for $x \ge 17$), the choice $t = c_4\, s + O(\log s)$, with explicit absolute constants, supplies at least $3\,(\omega(W) + \#E)$ primes, so $\Pr[\, p \in E \text{ or } p \mid W\,] \le 1/3$. If $p \in E$, Arthur accepts vacuously (inside the $1/3$). Otherwise he computes $W \bmod p$ in polynomial time: writing multiplication-by-β_i as 3×3 rational matrices M_{β_i} in the power basis, he evaluates $M_\beta \equiv M_{\beta_0} \prod_i M_{\beta_i}^{2^i} \pmod p$ by repeated squaring, then $W \equiv \mathrm{tr}(M_\beta M_\alpha) \bmod p$, accepting iff $W \equiv 0$.

Completeness. For a YES instance Merlin sends a true solution element: V1, V2 pass deterministically and $W = 0$, so $W \equiv 0$ modulo *every* prime — Arthur accepts with probability 1.

Soundness (one-sided). For a NO instance, any certificate passing V1 and V2 must have $z(\beta) \ne 0$ — otherwise β would be a solution — hence $W \ne 0$, and Arthur rejects with probability $\ge 2/3$. Arthur never rejects a valid certificate.

Amplification. k independent primes give error $(1/3)^k$; widening the range to 2^{s^c}, $c \ge 2$, already makes a single prime's error exponentially small, since $\omega(W)/\pi(2^{s^c}) \le 2^{c_4 s - s^c}\,\mathrm{poly}(s)$. □

Complexity-class bookkeeping. Step V3 alone is a coRP test: vanishing instances are accepted with probability 1, non-vanishing instances rejected with high probability. The outer existential quantifier over certificates therefore places the decision problem in $\exists\cdot\mathrm{coRP} = \mathrm{MA}$ — which is the precise sense of Proposition 3.1.

(Provenance: the notes' ledger listed this as "proposition-with-sketch, needs a careful write-up (height bound on $z(\beta)$ from the compact data; prime-range constants)". The bounds above discharge those two items; the residual dependence is the quoted format bounds of Section 2.2.)

4. The Compact Vanishing Problem

4.1. Statement and consequence.

CVP_K **(Compact Vanishing Problem).** Given a compact representation of an algebraic integer β in a fixed cubic field

K, decide in deterministic polynomial time whether a fixed $\mathbb{Q}$-linear coordinate functional vanishes at β.

For our instances the functional is c_2, the α^2-coordinate in $K = \mathbb{Q}(\sqrt[3]{d})$. (The acronym clash with the lattice Closest Vector Problem is noted; no lattice problem appears here.)

Corollary 4.1 (NP iff-modulo CVP)**.** *If* $\mathrm{CVP}_K \in \mathrm{P}$ *for the instances arising from Thue certificates, then pure cubic Thue solvability is in* NP*, unconditionally. Conversely — the "iff-modulo" gloss, architectural rather than formal — CVP is the* only *step of the canonical verifier that is not deterministic polynomial time: any* NP *proof through this certificate collapses to deciding CVP on these instances. Under standard derandomization hypotheses (which give* MA = NP*),* NP *membership holds conditionally today.*

The notes established three equivalent faces of CVP: a trace form, a lattice-membership form, a bounded-width branching-program form.

4.2. **Face one: the trace form.** For $\beta = x + y\alpha + z\alpha^2$ one computes $\beta\alpha = zd + x\alpha + y\alpha^2$, and $\mathrm{Tr}(1) = 3$, $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\alpha^2) = 0$ give

$$\mathrm{Tr}(\beta\alpha) \;=\; 3d\,z, \qquad \text{i.e.} \qquad z(\beta) \;=\; \frac{\mathrm{Tr}(\beta\alpha)}{3d}.$$

CVP for our functional is exactly: *is* $\mathrm{Tr}(\beta\alpha) = 0$ *for compact* β*?* The coordinate functionals *are* trace forms — the dual basis of the power basis under the trace pairing gives $c_2(\beta) = \mathrm{Tr}(w\beta)$ with $w = \alpha/(3d)$, and likewise for c_1, c_0 (identities machine-checked during this draft's assembly).

4.3. **Face two: plane membership for Arakelov-pinned elements.** The second face rests on a small clean result, the micro-yield of the notes' Ge-lattice attempt.

Lemma 4.3 (Arakelov pinning)**.** *From a valid compact representation of* $\beta \in O_K$*, a deterministic polynomial-time verifier computes the full* ***Arakelov divisor*** *of* β*: the exact ideal* $I = (\beta)$ *in Hermite normal form, and the log-embeddings* $\log|\sigma_j(\beta)|$ *to any requested* $\mathrm{poly}(s)$ *precision. This data determines* β *up to sign.*

Proof sketch. The ideal: clear the format's denominators first and work with the numerator ideal of the compacted element — so that I is integral, as the HNF intersection $L = I \cap M$ of Section 4.3 strictly requires — then reduce the power product by repeated squaring on reduced ideals with tracked distances — infrastructure arithmetic is precisely this machinery [BW88, Sch08] — every intermediate object staying poly-size. The logs: interval arithmetic on the factors' certified embedding data, via the exact telescoping of Section 3. Uniqueness: if $(\beta) = (\beta')$ with equal log vectors, β/β' is a unit with vanishing log-embeddings, hence torsion, hence ± 1 (the torsion of K is $\{\pm 1\}$ by the real embedding). *(sketch)* □

So the certificate pins a *unique* candidate (up to sign, harmless since $z(-\beta) = -z(\beta)$): **compact certificates verify everything about β except one plane membership.** Indeed $z(\beta) = 0 \iff \beta \in M$, and $L := I \cap M$ is an explicit rank-2 lattice with a poly-size basis (HNF intersection). Hence:

> **CVP (Thue instances) $\iff$ does the element pinned by a given Arakelov divisor lie in an explicit rank-2 sublattice?**

4.4. **Face three: width-3 ABP trace zero-testing.** Writing multiplication-by-β_i as 3×3 integer matrices (after clearing the format's denominators), the compact representation becomes a **matrix word with repeated-squaring structure**, and by the trace form CVP reads: *decide whether the trace of a width-3 integer matrix word is zero.* That is polynomial identity testing for width-3 algebraic branching programs — a highly structured, white-box, repeated-squaring instance, and the right literature door: it connects the NP question for cubic Thue equations to bounded-width PIT derandomization by the shortest path we know of (Section 5.4). One structural advantage must be stated explicitly: the matrices M_{β_i} all lie in the commutative subalgebra $\mathbb{Q}[M_\alpha] \cong K$ of 3×3 matrices — the word is not a generic non-commutative width-3 ABP but one over a **commutative** matrix ring, a class significantly better understood in the PIT literature and correspondingly a more tempting target for derandomization.

4.5. **Remark (Lagarias vacuity: why 1979 stopped at degree 2).**

Remark 4.5. In the quadratic case the module $\mathbb{Z} + \mathbb{Z}\sqrt{d}$ is *all* of $\mathbb{Z}[\sqrt{d}]$: the Thue-shape condition is vacuous, CVP never arises, and norm-checking alone suffices — which is why Lagarias's certificates [Lag79] close at degree 2 with purely multiplicative technology. The coordinate condition is genuinely **new at degree 3**: CVP is exactly what separates the cubic certificate problem from Pell — this, not the unit computation, is the honest frontier of the NP question, and the explanation of the shift from degree 2 (NP since 1979) to degree 3 (MA now).

5. Why the classical toolbox stops

We place CVP on the map, then close four attack routes — distilling the notes' 1991–2026 literature sweep and recorded attempts. No route is closed by a hardness theorem; each is closed by structural inapplicability, a different and more instructive thing.

5.0. **Where CVP sits: the sum-of-square-roots family.** The cleared trace value admits a poly(s)-size straight-line program (repeated squaring), so CVP is an instance of EquSLP — zero-testing SLP-given integers — in coRP (the random-prime test of Section 3; the observation goes back to Schönhage [Sch79]), with its positivity cousin PosSLP in the counting hierarchy [ABKM09, ESY14]. Deterministic polynomial time is open for the whole family, whose totem is the sum-of-square-roots problem: poly-time for explicitly given radical sums [Blo91], uniform TC^0 for deciding whether a rational radical sum is rational [HBMOW10], the separation-bound frontier only recently moved by the subspace theorem [EHS24]. The notes' verdict, adopted here: CVP is **genuinely new, of sum-of-square-roots-class flavor** — in coRP, in CH, deterministic case open — with one unexplored asset recorded in Section 7.2.

5.1. **The additive/multiplicative divide.** Thiel-style deterministic *equality* testing of compact representations is a **multiplicative** technology: $\beta = \gamma$ reduces to the quotient being 1, and units $\neq \pm 1$ are separated from 1 by Dobrowolski-type height lower bounds [Dob79] on the log-embedding lattice — polynomial precision suffices. Ge's theorem [Ge93] (see [vG25, Thm. 165]) extends this: *multiplicative relations* among power products are decidable in deterministic polynomial time — exactly why the norm step V2 is easy. The compact-representation toolbox stops at the trace, precisely: [BTW95] adds two-term sign comparison in real quadratic fields, and **no additive $\geq$ 3-term test exists anywhere in the literature**.

CVP is **additive**: $\mathrm{Tr}(\beta\alpha)$ is a sum of three conjugate terms. Separation is not the problem — integrality gives it for free (the cleared value is a rational integer, so nonzero means ≥ 1). **Evaluation is the problem**: certifying the sum numerically needs absolute precision comparable to the archimedean spread $\max_j \log|\sigma_j(\beta)| - \min_j \log|\sigma_j(\beta)|$, and for *true* Thue solutions the spread is $2^{\Theta(s)}$ forced: a solution has $|\sigma_1(\beta)| = |m|/|\sigma_2(\beta)|^2$ tiny *precisely because* it is a solution. The easy "balanced-β" case of CVP can therefore never contain the instances we care about. The angle formulation hits the same wall: verifying $\arg\sigma_2(\beta\alpha) \approx \pm\pi/2$ to error $e^{-\text{spread}}$ needs the small factors' arguments to exponentially many bits.

The conjugate-certificate attempt, and its circle. One can move to the Galois closure $L = K(\omega)$ and note $\mathrm{Tr}(\beta\alpha) = 0 \iff \beta + \omega\sigma_2(\beta) + \omega^2\sigma_3(\beta) = 0$; certifying compact representations of the conjugates is fine, but the final check is again an additive vanishing of compact objects — the divide reappears intact. **Sums of compact representations are not compact; that sentence is the obstacle.**

5.2. **Blömer is structurally inapplicable, not merely insufficient.** Blömer's deterministic polynomial-time algorithms for sums of radicals [Blo91, Blo93] might look like the natural import: our elements are $\mathbb{Q}$-combinations of $1, d^{1/3}, d^{2/3}$. But his determinism rides on *Siegel-type rigidity*: independent radicals admit **no** nontrivial $\mathbb{Q}$-linear relations, so zero-testing reduces to coefficient collection on explicit poly-size coefficients. Our trace-zero instances **are** nontrivial relations in the rank-2 trace-zero module — the exact phenomenon his theorem excludes — with coefficients x, y, z compact rather than explicit. Both pillars — rigidity and explicit arithmetic — fail simultaneously.

5.3. **No ideal-theoretic route: the hyperplane argument.** The tempting program "tear into Thiel's ideal equality tests to break CVP" is closed before it opens, by rank counting.

Proposition 5.1. *No ideal containment, equality, or divisibility statement in O_K can express the condition $z(\beta) = 0$, even in principle.*

Proof. The trace-zero condition $\mathrm{Tr}(\beta\alpha) = 0$ defines a $\mathbb{Q}$-**hyperplane** in K (the coordinate functionals are trace forms, Section 4.2). Ideals are **full-rank** $\mathbb{Z}$-modules: any nonzero ideal of O_K has rank 3, while any module contained in the hyperplane has rank ≤ 2. □

Thiel/Ge machinery tests relations among full-rank multiplicative objects; CVP lives in a corank-1 additive slice invisible to all of them. The bridge of Section 5.5 is not hiding inside ideal arithmetic; it must be genuinely new.

5.4. **The EquSLP caution, and the Baker route's ceiling.** Two doors from complexity theory, each honestly measured:

- **Iterated squaring is the hard core.** ABKM [ABKM09, Prop. 2.1] show that EquSLP's difficulty concentrates on repeated-squaring instances — our shape — and derandomizing such identity tests implies circuit lower bounds [KI04]. This is **a caution, not a hardness proof**, for our width-3 single words: CVP could be easy without lower-bound consequences, but the precedent says the general-toolbox route is priced.
- **The Baker route has a height ceiling.** Galby–Ouaknine–Worrell [GOW15] decide sign and zero questions for entries of M^n, dim ≤ 3, in polynomial time — for **poly-height** inputs: the engine is lower bounds for linear forms in logarithms [Mat00], whose constants stay polynomial only while heights and the number of logarithms stay bounded. Thue certificates have $h(\varepsilon) = R/3 = 2^{\Theta(s)}$; at $\Theta(s)$ logarithms the Baker constants degrade exponentially — the same wall that pins LRS Positivity at order ≥ 6 [OW14]. The bounded fragment this route *does* give is Proposition 6.1.

5.5. **One wall, four costumes; the missing tool, named.** The four failures above are one wall:

(1) **Trace non-multiplicativity** (Section 5.1).
(2) **Sums of compacts are not compact** (Section 5.1).
(3) **The archimedean spread** (Section 5.1): $2^{\Theta(s)}$ precision forced exactly on true solutions.
(4) **Plane non-stability** (the notes' Ge-lattice attempt): the rescaling trick — bring the thin strip $\{|x - y\alpha| \sim e^{\ell_1}\}$ to bounded size by ε^{-j} — dies because M is not stable under units: $\varepsilon^{-j}M \neq M$. Lattice-point location in the plane is polynomial *at polynomial scales*, but our scales are $e^{2^{O(s)}}$, described only by their logs.

What a solution needs, stated once: a **multiplicative-to-additive bridge** — any theorem letting membership in a fixed rank-2 plane be tested through data carried multiplicatively (relation lattices, Arakelov classes, unit orbits). Nothing in the 1991–2026 literature provides one; building it is the actual open problem, and by Section 4.3 it now has a precise geometric formulation.

6. Positive fragments

6.1. The bounded fragment: CVP at bounded rank and small height.

Proposition 6.1 (bounded CVP — stated with a proof plan)**.** *Restrict* CVP_K *to instances whose factor list* $\{\beta_0, \dots, \beta_T\}$ *generates a subgroup of* $K^\times/\{\pm 1\}$ *of multiplicative rank* $O(1)$ *admitting generators of Weil height* $\mathrm{poly}(s)$*. This fragment is decidable in deterministic polynomial time.*

Proof plan (honestly flagged: a plan, not a proof). (1) Compute the relation lattice of the factors — deterministic polynomial time by Ge [Ge93], [vG25, Thm. 165] — and rewrite the word over $O(1)$ independent poly-height generators with binary exponents. (2) The trace becomes a fixed short sum of conjugate power products in $O(1)$ poly-height bases. (3) Zero-test via the Baker–Matveev machinery as in the dimension-3 matrix-power analysis of [GOW15]: with $O(1)$ logarithms of poly-height numbers and binary exponents, Matveev's bounds [Mat00] are polynomial — noting that Matveev's constant degrades exponentially in the number k of logarithms, which is exactly why the $O(1)$ multiplicative-rank restriction is load-bearing and cannot be relaxed, so certified interval evaluation at $\mathrm{poly}(s)$ precision decides against the free integrality separation. Steps (2)–(3) need the same case-discipline GOW exercise at dimension 3 and are not written out here. ☐

What this fragment is and is not. It is the notes' verdict "publishable-but-bounded". It does **not** cover Thue certificates: their factor lists contain ε (or its walk factors), of height $R/3 = 2^{\Theta(s)}$ — precisely the ceiling of Section 5.4. No claim of progress on the Thue NP question is made here.

6.2. The strong window conjecture: prover efficiency and coNP.

The strong window conjecture [companion, Remark 7.4] — every Thue-shape index of a unit-reduced representative satisfies $|k| \le C(1+\log|m|/R)$, hence $O(s)$ by the regulator floor [ADF16] — turned out to be unnecessary for YES-certificates (Section 2.3). Its remaining roles: **prover efficiency** — the honest prover runs the companion's algorithm once ($2^{O(s)}$); the conjecture caps the final scan at $O(s)$ indices per representative, though the infrastructure walk remains $\widetilde{\Theta}(R)$, so it trims the narrative, not the exponential — and **the coNP side**, where certifying a NO instance would require certifying every ideal class's orbit window zero-free, and the conjecture caps the indices to certify per orbit at $\mathrm{poly}(s)$, making class-enumeration NO-certificates conceivable. *(Ledger flag, preserved: the coNP side is untouched in this work.)*

7. Open problems

7.1. **CVP itself.** Is CVP_K — even just for Thue-shaped instances — in P? The notes' two probed routes are blocked, not dead: the *p-adic route* (ord_p of power products is poly-computable per prime) stumbles on the fact that valuations do not see addition; the *canonical-form route* (Thiel equality of β against its projection) needs a compact representation of the projection, and sums of compacts are not compact. The sharpest open formulation is the ABP face: **is the repeated-squaring width-3 matrix-word class among the deterministically testable bounded-width PIT cases?** A positive answer completes: cubic Thue $\in$ NP, unconditionally — Lagarias one rung up, for real.

7.2. **The multiplicative-to-additive bridge.** Build any theorem letting membership in a fixed rank-2 plane be tested through multiplicatively carried data (Section 5.5). The one unexplored asset, recorded in the notes' literature verdict: our three summands are Galois-conjugate, and the bases' full multiplicative relation lattice is *computable* (Ge). The additive analog of [vG25, Lemma 166] — detecting trace-zero from the relation lattice plus unit-lattice geometry — has never been attempted. That is the door.

7.3. **DIVIS hardness.** The program's synthesis records that the divisibility language $\mathrm{DIVIS} = \{(A, B) \in \mathbb{Z}[x]^2 : \exists x \in \mathbb{Z},\ A(x) \mid B(x)\}$ is in NP (two proofs, one the expedition's, by an elementary integer-remainder argument — Runge in miniature); its hardness is unstudied. It is the nearest candidate for a hardness foothold adjacent to the Thue stratum.

7.4. **The coNP certificate question.** Is pure cubic Thue solvability in coNP? Degree 2 says yes one rung down (negative Pell is in NP $\cap$ coNP [Lag79]). At degree 3 the ingredients on the table are the strong window conjecture and class-enumeration certificates (Section 6.2); nothing beyond the framing exists yet.

Status ledger (carried from the notes, unchanged in spirit).

- MA membership: proposition with the height-bound and prime-range write-up supplied (Section 3); the format-bound dependence is now DISCHARGED by the pinned citations (Section 2.2).
- Compact-representation existence and size: **debt paid** — Thiel's thesis read in full (Def. 6.3.1, Cor. 6.3.9–6.3.10, unconditional, arbitrary degree), efficient computation pinned to [BF14, Alg. 7, Prop. 5.1]; the erroneous compaction citation to [Sch08] corrected (it contains no compact representations — verified against the source). The remaining ledger flags may now be stripped at final submission.
- CVP: open, named, isolated — the right-sized next mathematics.
- coNP side: untouched here.

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